

A heat recovery rate of the thermochemical waste-heat recuperation systems based on experimental prediction



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ABSTRACT

The concept of the thermochemical waste-heat recuperation system by steam methane reforming is considered. The heat recovery rate of the recuperation system can be considered as the main criterion of its efficiency. The methodology of determination of the recovery rate of the thermochemical recuperation system is developed. The recovered heat in such systems is a sum of the recovered heat in a steam generator and a reformer. The experimental investigation was performed from 500 to 1000°C and for various steam-to-methane ratio from 1 to 3. The total recovered heat and the recovered heat in the steam generator and the reformer are calculated. In the temperature range below 700°C the total recovered heat increases significantly due to the sharp increase in the enthalpy of steam methane reforming reaction. In the temperature range above 700°C, the increase in the total recovered heat is carried out mainly due to the preheating of the synthesis gas since the reaction enthalpy reaches maximum value. The heat recovery rate of the thermochemical recuperation system is determined. It is established that the maximum recovery rate is for 700°C and $H_2O:CH_4 = 2$. At temperature of 700°C, the thermochemical recuperation system is capable of recovering more than 85% heat of exhaust gases.

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1. Introduction

The industrial sector widely consumes natural gas as a fuel for heat generation in the heat-power and heat-technology systems. In the last years, these systems accounted for more than 60% of natural gas consumption in the world [1]. Despite great progress in the efficient use of natural gas in the heat-power and heat-technology systems, there is still a great potential for increasing energy efficiency of such systems [2].

There are two main ways to improve the fuel efficiency of the fuel-consuming equipment in the heat-power and heat-technology systems [3]. The first is the rational use of traditional fuel through the improvement of fuel and technological cycles. The second is the replacement of the original fuel with alternative fuel and energy resources. A special place among alternative hydrocarbon fuels is (in view of high environmental friendliness) the so-called synthetic fuel (synthesis gas) - hydrogen-containing gas that obtained directly in the fuel-consuming system. Such technology of synthetic fuel generation is sometimes called as on-board hydrogen production [4–7]. One of the ways to increase energy efficiency,

which satisfies the above two directions, is thermochemical waste-heat recuperation [8–13].

Recently scientists and engineers from different countries have published a series of articles on thermochemical recuperation. There are papers on the experimental [14–16] and theoretical study [10,12,17] of the thermochemical waste-heat recuperations in various fuel-consuming equipment: the internal combustion engines [11,18], the gas turbines [10,19], the industrial furnaces [9,12,16], etc. The variety of such articles shows the promise of this way of the energy efficiency increasing of fuel-consuming equipment as well as an environmental performances of such equipment [20].

The use of thermochemical waste-heat recuperation (hereinafter TCR) in the industrial plants has been widely discussed for high-temperature plants, in particular, heat-treating and glass melting furnaces. Rue and co-authors [12] showed the results of testing of a glass melting furnace with TCR in a pilot scale process unit. The authors showed that in a furnace with TCR part of waste heat is used for the reforming of natural gas. They concluded that simulation and experimental investigation in this paper have been shown that energy recovery in the TCR systems is notably higher than it is possible recuperation with either thermal regenerators or heat exchangers [12]. For example, fuel saving for the furnace with

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TCR was 25–30% higher relative to a furnace with the recuperation systems based on air preheating.

In addition, Popov and co-workers [9] published the results of a thermodynamic study of the industrial plants with thermochemical recuperation by methane reforming. They also considered the glass melting furnace with TCR. The authors analysed the glass melting furnaces with various TCR configuration via special software such as Mathcad and Aspen Plus. Popov et al. [9] concluded the TCR can give up to 10–25% of energy efficiency increase in comparison with the conventional recuperation systems.

Continuing the topic of using TCR in the industrial furnaces, there is a series of articles that discuss the use of TCR in steel-making industry [16], in particular, energy recuperation from high-temperature slags by steam methane reforming [21,22], etc. In addition, Kobayashi patented several schemes for the use of TCR by natural gas reforming [23,24]. Scientists and engineers from different countries have tested and experimentally investigated the TCR systems based on fuel reforming for various industrial furnaces over the last several decades [25,26].

The use of thermochemical exhaust heat recuperation in gas turbine and combined-cycle plants has also found a wide discussion in the scientific literature. Kesser et al. [27] investigated chemically recuperated gas turbine as a respective way to obtain ultra-low NOx emissions. Kravchenko and Verkhivker [10] calculated that the total energy efficiency of a combined cycle plant with TCR has obtained about 80–90%. Both the works mentioned above [10,27] deal with TCR by steam methane reforming. However, Cao and co-authors [28] presented the results of a thermodynamic study of a gas turbine plant with TCR by dry-methane reforming. They reported that the optimized net power generation efficiency of the cycle rises up to 49.6% and the exergy efficiency rises up to 47.9%.

At the end of the introduction, it is necessary to mention the use of TCR in internal combustion engines. It is worth noting that the idea to use fuel reforming was first put forward by Lee and Wimmer [29] in the 1960s. The main aim of this investigation was focused on reducing pollutant emissions when a product gas (gas after endothermic transformation) was used as the main fuel. The historical retrospective of the use of the fuel reforming in the ICE is comprehensively presented by Tartakovsky and Steintuch [11]. Moreover, there are a set of articles presented by Tartakovsky and co-authors, where the results of using TCR by alcohols reforming are reported [30–32]. There is an article discusses the concept of a direct-injection ICE (internal combustion engine) with thermochemical heat recuperation by methanol steam reforming [32]. Poran and Tartakovsky reported that a possibility of efficiency

improvement of ICE by 12–14% compared with a DI-gasoline-fueled counterpart by applying a DI-ICE with thermochemical recuperation by methanol reforming at high pressure.

Moreover the thermochemical fuel transformation processes found a wide application for the solar thermochemical fuel production systems integrated with fossil fuel recuperation system [33]. In such systems, thermochemical cycling (TC) technology is applied. TC is a technology for solar energy storage via the synthesis of chemical fuels by utilizing concentrated solar heat alone to drive endothermic chemical reactions [34,35].

The main criterion for the effectiveness of the TCR system is the heat recovery rate of exhaust heat. The recovery rate shows how much the exhaust heat is usefully returned to the fuel cycle in relation to the total exhaust heat. Despite the variety of articles dealing with TCR, there is no single and universal method for determining the heat recovery rate. This is due to several reasons. First, various chemical endothermic reactions are used for TCR. Secondly, various schemes TCR is applied. Thirdly, various designs and characteristics of fuel reformers are used.

The main goal of the presented article is the development and scientific justification of a universal method for determining the heat recovery rate of the exhaust heat in the TCR systems. In addition, based on the developed method and previous author's experimental studies, the efficiency of the TCR system by steam methane reforming with point of view to the recuperation rate was determined.

2. Thermochemical recuperation concept

The general idea of thermochemical recuperation is to use the exhaust heat for fuel reforming of initial fuel. As a result of this transformation, new synthetic fuel (product of fuel reforming) has higher low heat value (LHV) in comparison with LHV of initial fuel. Endothermic reactions for TCR may be different. However, steam methane reforming has a wide application for thermochemical recuperation of high temperature exhaust heat [8–10].

A schematic diagram of a gas furnace with thermochemical recuperation by steam methane reforming is shown in Fig. 1. The schematic diagram can be virtually divided into two blocks: (1) a thermochemical recuperation system (a methane reformer, a steam generator (boiler), a mixer (steam ejector)); (2) a fuel-consuming system (fuel-consuming equipment itself, burner assembly).

In this schematic diagram, the waste flue gas after a gas furnace is sequentially passing through the methane reformer and a boiler (steam generator). The methane reformer performs as a catalytic heat exchanger [36–38]. The reaction mixture before the reformer

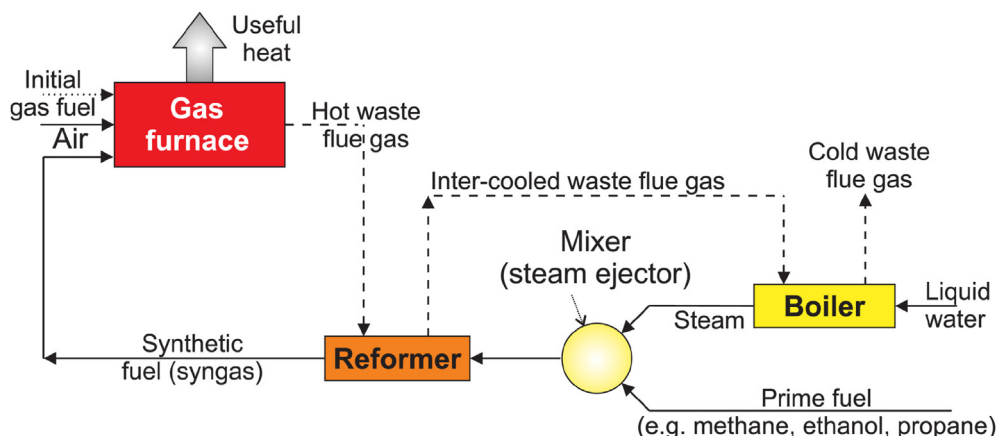
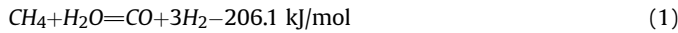


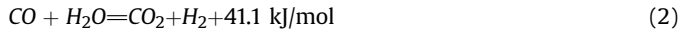
Fig. 1. Schematic view a gas furnace with thermochemical recuperation by steam methane reforming.

is containing steam and methane. Inside the reaction space of the reformer, methane and steam are reacted and the mixture that mainly consists of hydrogen is produced. The chemical endothermic reaction of the steam methane reforming process can be written as follows [39–41]:

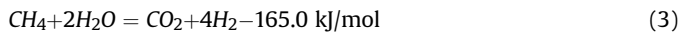
- steam methane reforming reaction:



- water-gas shift reaction:



- global steam methane reforming reaction:



In a conventional gas furnace, fuel is burned directly in a combustion chamber. For example, when combustion of 1 m³ of methane takes place, 35.82 MJ is produced in a combustion chamber. In the gas furnaces with thermochemical recuperation, original fuels are initially converted to new synthetic fuel (synfuel). For example, when burning synfuel produced by full reforming of 1 m³ of CH₄, 45.03 MJ is produced. In summary, it can be concluded that in the furnace with the TCR system the heat production from the natural gas is split into two blocks. The first block is the increasing of the LHV of the initial fuel by the transformation of exhaust heat (H_{exh}) into the chemical energy of new synthetic fuel. The second block is the combustion of this synthetic fuel that has higher LHV than that of the original fuel. The methods of heat extraction from the original fuel for the conventional gas furnace (a) and for the gas furnace with TCR by steam methane reforming (b) are presented in Fig. 2.

The main criterion for the energy efficiency of the heat recuperation system is the recovery rate [42,43]. The heat recovery rate shows how much heat of the exhaust gases is recovered in the fuel cycle of a gas furnace. Obviously, the efficiency of using the TCR systems is determined by the conversion of original fuel (degree of methane conversion). If the reforming reaction does not proceed, the efficiency of the TCR system is zero. On the other hand, the methane conversion depends on operating temperature, pressure and methane/steam feed composition, and the reformer characteristics [44–46]. Moreover, an efficiency of TCR application depends also on the rate of coke formation. The latter accumulates on the catalyst and leads to deterioration of primary fuel conversion.

Therefore, the heat recovery rate depends on operating conditions and the design characteristics of the reformer.

3. Methodology

The efficiency of application the TCR systems by steam methane reforming is depending on the heat transferred in the reformer for the methane reforming reaction and syngas heating. If steam methane reforming doesn't occur in the reformer, then the efficiency of using of the TCR system equals to zero or even has a negative effect, due to the fact that exhaust heat is used for generation of steam. In addition, an increasing steam-to-methane ratio in feed stream leads to an increase of mole fraction of non-combustible species in syngas, in particular steam. Residual steam and carbon dioxide (produced by water-gas shift reaction) are the components of syngas but from the energy point of view, residual steam and CO₂ doesn't play a positive role in the combustion process, because H₂O and CO₂ is non-combustible inert material of synfuel. Moreover, a large amount of heat is required for steam generation and from an energy point of view, a large molar fraction of steam in synfuel is negative [19]. Therefore, determining the optimal steam-to-methane ratio in the TCR system is an important engineering task.

It is known to prevent the formation of carbon in the reformer, the steam-to-methane ratio is chosen more stoichiometric [47,48]. In this regard, steam will be present in synthesis gas. The heat balance of a combustion chamber of an industrial furnace (for complete combustion) is as follows:

$$LHV_{\text{sg}} + H_{\text{air}} + H_{\text{sg}} = Q_{\text{use}} + H_{\text{exh}} + Q_{\text{loss}} \quad (4)$$

Where LHV_{sg} is total heat combustion of syngas produces by reforming of 1 mol of CH₄, kJ; H_{air} , H_{sg} is total enthalpy of air and syngas for combustion, respectively, kJ; Q_{use} is useful heat, kJ; H_{exh} is enthalpy of exhaust gas, kJ; Q_{loss} is other losses, kJ.

The enthalpy of synthesis gas is determined by the enthalpy of each component, including enthalpy of H₂O and CO₂. However, can enthalpy of H₂O and CO₂ be considered as a recovered heat? Obviously not. In this case, the feeding of H₂O and CO₂ will be similar to the scheme with flue gas recirculation [49–51]. In the recuperation systems based on air and fuel heating, a recovered heat is considered as the heat for preheating of combustion components (air and fuel). It is logical to assume that in the TCR systems, the recovered heat must also be counted through the enthalpy of the only combustion components, taking into account the enthalpy of the chemical reaction. The combustion components of synthesis gas are H₂, CO, residual CH₄. Although H₂O and CO₂ are part of synthesis gas, these components do not participate in a combustion process.

According to the definition, the heat recovery rate can be

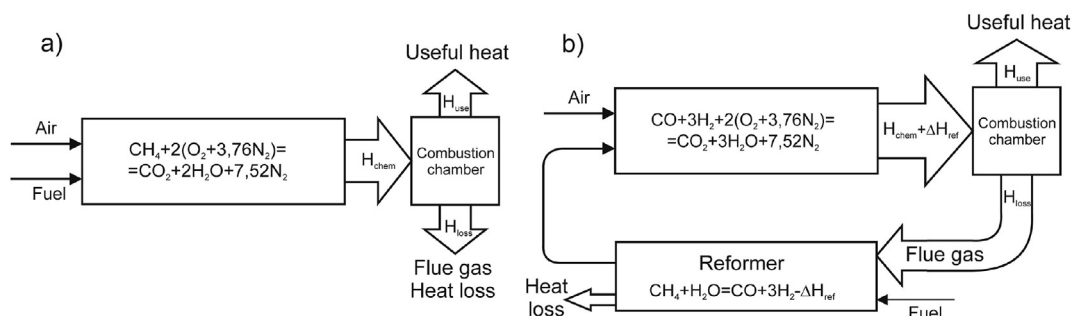


Fig. 2. The methods of heat generation from the original fuel for a conventional gas furnace (a) and for a gas furnace with TCR (b).

expressed as follows:

$$R = \frac{H_{rec}}{H_{exh}} = \frac{H_{rec}^{ref} + H_{rec}^{st.gen}}{H_{exh}}, \quad (5)$$

where H_{rec} — a recovered heat in the TCR system, kJ; H_{rec}^{ref} , $H_{rec}^{st.gen}$ — a recovered heat in a reformer and a steam generator, respectively; H_{exh} — an exhaust heat, kJ.

The enthalpy of the exhaust gases depends on the steam-to-methane ratio in the reformer. The enthalpy of the exhaust gases for 1 nm³ of inlet methane (at excess air ratio equal to unity) can be determined as follows:

$$H_{exh} = [n_{CO_2} \cdot c_{\mu, CO_2} + n_{N_2} \cdot c_{\mu, N_2} + (n_{H_2O} + \beta) \cdot c_{\mu, H_2O}] \cdot T_{exh}, \quad (6)$$

Where n_i is the number of moles of i -th element of the combustion products, mol; $c_{\mu, i}$ is the molar heat capacity, J/(mol·°C); $\beta = H_2O^{in}/CH_4^{in}$ is steam-to-methane ratio at the reformer inlet.

According to Fig. 1, the exhaust gases are used in the reformer for the steam methane reforming reaction and for heating of syngas. The heat transferred to the reformer from hot waste flue gas can be calculated as follows:

$$\Delta Q_{reformer} = H_{exh} - H'_{exh} \neq H_{rec}^{ref}, \quad (7)$$

where H'_{exh} is enthalpy of inter-cooled exhaust gas (according to Fig. 1).

The heat transferred to the reformer is not equal to the recovered heat because syngas consists of combustible elements (H₂, CO, CH₄) and non-combustible elements (H₂O, CO₂). Therefore, the enthalpy of noncombustible elements in syngas can't be taken into account as a recovered heat because in this case, these elements are the inertial materials. Therefore, the recovered heat in the reformer can be calculated as follows:

$$H_{rec}^{ref} = H_{smr} + H_{sg} - H_{H_2O} - H_{CO_2} - H_{mix}, \quad (8)$$

where H_{smr} — enthalpy of steam methane reforming reaction, kJ; H_{sg} — syngas enthalpy, kJ; H_{H_2O} , H_{CO_2} — enthalpy of steam and carbon dioxide that are contained in syngas, kJ; H_{mix} — enthalpy of steam-methane mixture before the reformer, kJ.

In expression (8), the term of ($H_{sg} - H_{H_2O} - H_{CO_2} - H_{mix}$) shows the recovered heat in the reformer by syngas heating (H_{rec}^{syngas}).

Synthesis gas enthalpy, the enthalpy of H₂O and CO₂ in synthesis gas, as well as the reaction enthalpy depends on the reaction conditions. The composition of synthesis gas after the reformer can be determined both experimentally [38] and theoretically [52]. The reaction enthalpy is difficult to directly determine. However, on the basis of thermodynamic analysis, it can be assumed that at the high temperatures, the main contribution to the material and heat balance is made by reaction (1) in comparison with the water-gas shift reaction (2). Therefore, we can assume that the reaction enthalpy is determined by a methane conversion:

$$H_{smr} \approx X_{CH_4} \cdot \Delta H_{298}^0; \Delta H_{298}^0 = 206.1 \text{ kJ/mol}, \quad (9)$$

where X_{CH_4} is a methane conversion.

In the expression for the heat recovery rate (5), there is the term of recovered heat in the steam generator. The heat of inter-cooled exhaust gas that transferred to steam generator also cannot be considered as a recovered heat because part of steam (if steam-to-methane ratio greater than 1 or if CH₄ conversion is less 1 for steam-to-methane ratio of 1) is containing in syngas and also don't play a role (from an energetic point of view) as a combustion component:

$$\Delta Q_{st.gen} = H'_{exh} - H''_{exh} \neq H_{rec}^{st.gen}, \quad (10)$$

where H''_{exh} is enthalpy of cold exhaust gas, kJ.

Therefore, $H_{rec}^{st.gen}$ must be calculated as follows:

$$H_{rec}^{st.gen} = (h_{steam} - c_p \cdot t_w^{in}) (C_{steam}^{in} - G_{steam}^{sg}), \quad (11)$$

where h_{steam} is the enthalpy of steam after a steam generator, kJ; C_{steam}^{in} is the mass flow rate of water before a steam generator, kg/s; G_{steam}^{sg} is the mass flow rate of steam in syngas, kg/s; t_w^{in} is temperature of feed water, °C; c_p is the heat capacity of feed water, kJ/(kg·°C).

Schematically, the algorithm of the recovery rate calculation for the TCR systems by steam methane reforming is presented in Fig. 3.

The main task for determining the heat recovery rate of the TCR

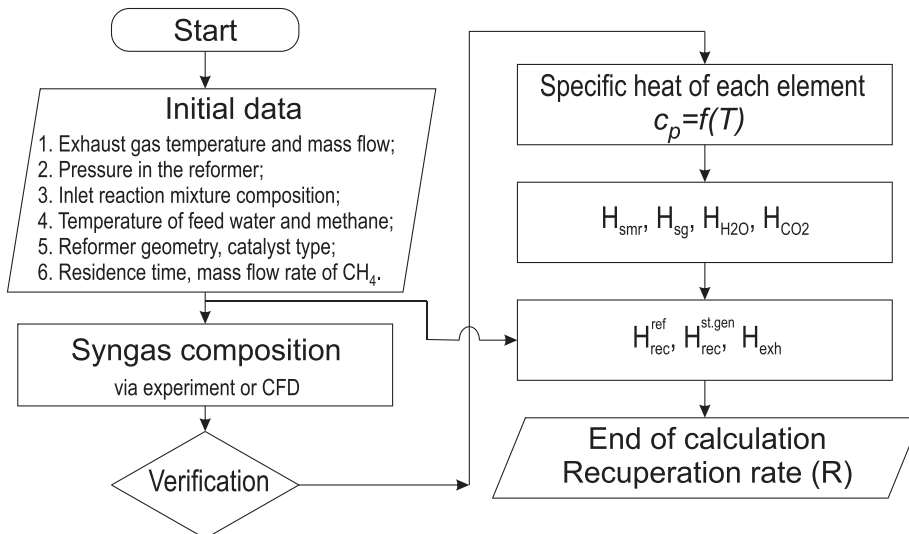


Fig. 3. An algorithm of a heat recovery rate calculation for the TCR systems.

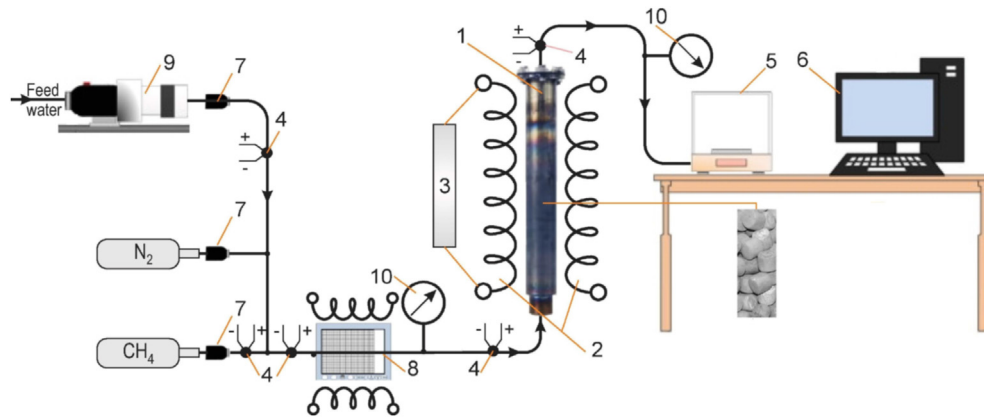


Fig. 4. Schematic view of experimental setup [37].

systems is the determination of syngas composition after the reformer. There are two main ways to determine syngas composition after the reformer. The first way is an experimental method; the second way is based on CFD-modelling [45,53,54]. It is known that the composition of synthesis gas depends on the operating parameters, such as temperature, pressure and composition of the feed mixture, on the residence time (contact time of reaction mixture with a catalyst), on the characteristics of the reformer such as the composition and shape of the catalyst, the size of the reaction space.

In this study, syngas composition for calculating the heat recovery rate is partly taken from the results of the author's previous experimental studies. The author's previous works deal with an experimental study of a methane reforming process [36,37]. The experimental setup for an investigation of steam methane reforming, the catalysts, the experimental procedure, etc are described in details in the previous author's articles [37]. The experimental setup in Fig. 4 mainly consists of the following elements: 1 – a methane reformer; 2 – an electrical muffle furnace; 3 – a watt-hour meter; 4 – the thermocouples; 5 – a portable gas analyzer; 6 – a personal computer; 7 – the mass-flow meters; 8 – an electrical preheater; 9 – an electrical steam generator; 10 – a manometer.

In this study, the author does not set the main goal to show the results of an experimental study of the steam methane reforming process. The main goal of the presented work is to calculate the recovery rate on the basis of the experimental results. For the experiments, the commercial Ni-based catalyst was used. The shape of catalyst is a cylinder with height of 15 mm and diameter 15 mm.

During the experiments, it was found that synthesis gas composition at the reformer outlet is a close to equilibrium composition at high residence time ($> 5.0 \text{ kg}_{\text{cat}} \cdot \text{s} / \text{mol}_{\text{CH}_4}$), which corresponds to a volume flow rate of 4000 h^{-1} . This fact is confirmed by the works of other authors [14].

4. Results and discussion

4.1. Syngas composition

The gas analyzer of the experimental setup allows determining the content of methane, hydrogen, carbon monoxide, carbon dioxide and steam in synthesis gas. The duration of each experiment was 10–15 min. Therefore there was no noticeable carbon content in the reaction products. In real conditions with long-term operation on the surface of the catalyst, carbon deposition is possible with a low steam-to-methane ratio. A fresh catalyst was used for

Table 1

The composition of synthesis gas for different steam-to-methane ratios at $p = 3 \text{ bar}$.

Temperature, °C	β	H ₂	CO	CO ₂	H ₂ O	CH ₄
500	1.0	0.33	0.03	0.08	0.25	0.31
	2.0	0.35	0.01	0.08	0.37	0.19
	3.0	0.33	0.01	0.07	0.51	0.08
600	1.0	0.52	0.1	0.07	0.13	0.18
	2.0	0.52	0.08	0.08	0.24	0.09
	3.0	0.45	0.06	0.06	0.37	0.05
700	1.0	0.65	0.19	0.03	0.06	0.07
	2.0	0.63	0.13	0.06	0.16	0.02
	3.0	0.54	0.11	0.05	0.30	0.00
800	1.0	0.72	0.22	0.01	0.02	0.03
	2.0	0.65	0.15	0.04	0.16	0.00
	3.0	0.54	0.13	0.04	0.29	0.00
900	1.0	0.73	0.25	0.01	0.01	0.02
	2.0	0.64	0.16	0.04	0.17	0.00
	3.0	0.53	0.13	0.03	0.30	0.00
1000	1.0	0.72	0.26	0.00	0.01	0.01
	2.0	0.64	0.16	0.03	0.17	0.00
	3.0	0.53	0.13	0.03	0.31	0.00

each experiment. Table 1 shows the composition of the synthesis gas for different steam-to-methane ratios. The steam-to-methane ratio is denoted as β .

4.2. Source data

Before proceeding to determine the heat recovery rate, it is necessary to give some supplements and simplifications. In a real TCR system (Fig. 1), the reaction mixture is heated in a reformer. In a laboratory setup (Fig. 4), the reaction mixture before the reformer is heated by an electric heater. In addition, in a real TCR system, the reformer is heated by exhaust flue gases, in contrast to the laboratory setup. Moreover, in a real TCR system, the temperature of the synthetic gas at the outlet of the reformer depends on the design and size of the reformer. In order to perform a correct comparison of the heat recovery rate for different operating parameters, it is assumed that the temperature of the synthesis gas is 100°C lower than the temperature of the flue gases. This assumption is based on experimental data. The synthesis gas composition is determined by the syngas temperature at the reformer outlet. Such an assumption may give a slight error. However, this assumption will make it possible to correctly compare the TCR systems that operate in different conditions.

Moreover, in the laboratory setup (Fig. 4), steam is at saturation. In a real TCR system, it is advisable to use superheated steam, since in this case, the initial temperature of the reaction mixture

Table 2

The initial data for calculation of a heat recovery rate.

Parameter	Pos. in Fig. 1	Value	Unit
Temperature	Feed water	70	°C
	CH ₄ inlet	20	°C
	Synfuel	$T_{exh}-100$	°C
	Reformer	$T_{exh}-100$	°C
	Steam	$f(p_s)$	°C
Volume flow rate	CH ₄	1	nm ³
Pressure	Boiler, reformer	equal	Pa

increases. For the calculations of the recovery rate, it is considered that the steam after the steam generator is at saturation. In addition, in a real TCR system, heat from flue gases is used to produce steam. In the laboratory setup the steam is generated by an electric steam generator. At a high steam-to-methane ratio, heat deficit in the TCR system is possible, therefore, external sources of heat must be used to generate steam. The detailed discussion on a heat deficit in the TCR system are presented below.

In addition, in order to compare the heat flows at different operating parameters (temperature, pressure, and inlet steam-ethanol composition), the following initial data were taken. The initial data are presented in Table 2. There are in Table 2 T_{exh} —temperature of exhaust gases after gas furnace and p_s —saturation pressure.

4.3. Heat flows

According to the methodology for determining the recovery rate in Section 3, the recovered heat in the TCR system is the sum of the recovered heat in the steam generator and the reformer. In addition, the recovered heat in the reformer is the sum of the enthalpy of the reforming process and the increase in the enthalpy of synthesis gas due to reheating in the reformer. Fig. 5 shows the enthalpy of the reforming process, the recovered heat in the steam generator and the recovered heat by heating the synthesis gas in the reformer as a function of an exhaust gas temperature. The data from Fig. 5 is calculated based on the experimental results presented in Table 1. All calculations are for 1 normal m³ (1 nm³) of methane. Fig. 5a and b shows the recovered heat for H₂O:CH₄ = 1:1 and H₂O:CH₄ = 2:1, respectively.

From Fig. 5, it can be seen that as the temperature increases, the enthalpy of the steam reforming process increases. The enthalpy of steam methane reforming shows how much flue gas heat has been transformed into the chemical energy of new synthetic fuel. In

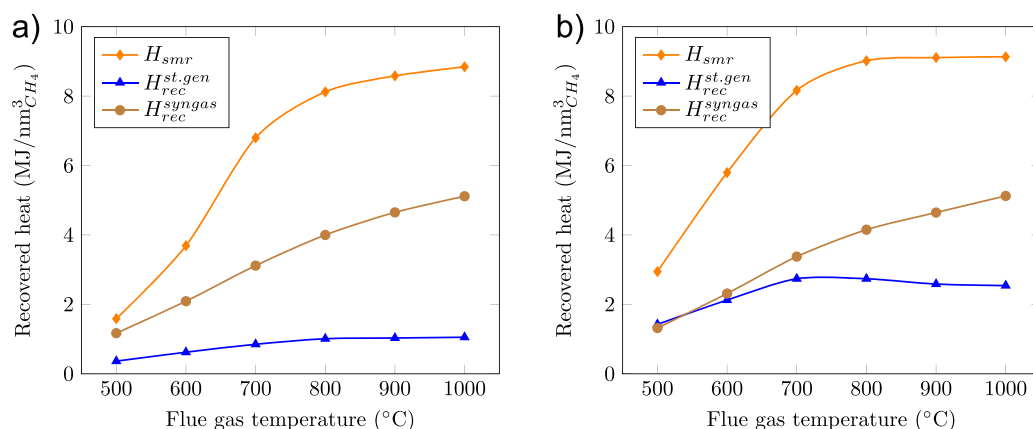
other words, the enthalpy of the SMR process shows how much the LHV of new synthetic fuel is higher than the LHV of the original methane. The character of the H_{smr} curves in Fig. 5, which were obtained using experimental data, is a good agreement with the nature of the enthalpy curve of the SMR reaction, obtained using thermodynamic analysis [8]. The increase in H_{smr} in the temperature range up to 750°C has a sharp character. Further, increase H_{smr} slows down. This is due to the fact that in the temperature range above 750°C the reversible reaction (1) basically is a direct reaction. This fact is characterized by a large value of the equilibrium constant in this temperature range.

Fig. 5 shows that the increase in H_{rec}^{syngas} has a smooth character. This is due to the fact that the change in the enthalpy of synthesis gas mainly occurs due to an increase in temperature, and the composition of the synthesis gas affects enthalpy slightly. Moreover, Fig. 5 shows that an increase in the steam-to-methane ratio leads to a marked increase in recovered heat in the steam generator. This is due to the fact that when the steam-to-methane ratio increases the steam consumption for the steam methane reforming process is increasing. In the temperature range up to 700°C, steam consumption occurs in the steam reforming reaction and in the water gas-shift reaction. When the temperature rises above 700°C, steam consumption occurs only in the steam reforming reaction, since water gas-shift reaction (2) has a reverse direction in this temperature range.

The total recovered heat of the TCR system as a function of exhaust gas temperature for the various steam-to-methane ratios ($\beta = \text{H}_2\text{O} : \text{CH}_4$) is presented in Fig. 6. Fig. 6 has a dashed line for $\beta = 3$ that shows the total recovered heat, taking into account the heat deficit. The heat deficit is discussed in several paragraphs below. In the temperature range below 700°C the total recovered heat increases significantly due to the sharp increase in H_{smr} . In the temperature range above 700°C, the increase in the total recovered heat is carried out mainly due to the preheating of the synthesis gas since H_{smr} reaches maximum value.

Based on the methodology described in Section 3, the recovery rate of the thermochemical waste-heat recuperation system is determined. Fig. 7 shows the dependence of the recovery rate on the temperature of the flue gases for different steam-to-methane ratios.

Fig. 7 clearly shows that the efficiency of a TCR system greatly depends on the temperature of the exhaust gas. As mentioned above, the basic concept of the TCR systems is the transformation of the exhaust heat into the chemical energy of new synthetic fuel. The result of this transformation is the increased LHV of synfuel

**Fig. 5.** The recovered heat in the reformer and steam generator.

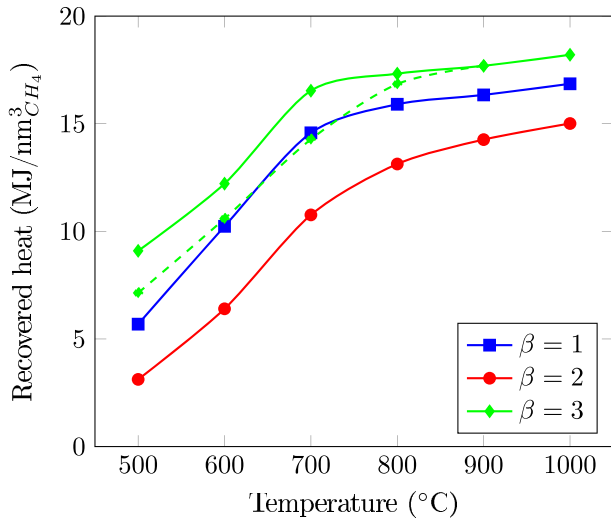


Fig. 6. A total recovered heat of a TCR system as a function of exhaust gases temperature.

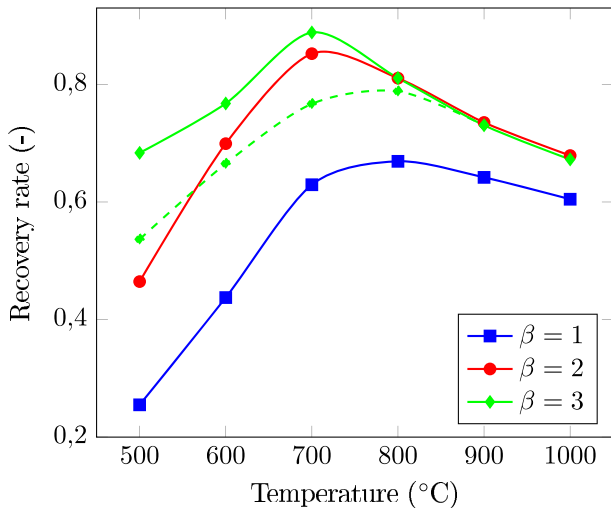


Fig. 7. The heat recovery rate of the TCR system vs. temperature of the exhaust gases.

compared to the LHV of methane. That is why when the enthalpy of the steam reforming reaction increases, the efficiency of the TCR system increases. Fig. 7 shows that the maximum heat recovery rate takes place at 700°C and $\beta = 2$.

Fig. 7 shows that after reaching the maximum value in the temperature range of about 700°C, the heat recovery rate decreases with increasing temperature. This is due to two main reasons. First, as the exhaust gas temperature increases, the denominator in expression (5) increases. Secondly, for the ratios $\beta = 2$ and $\beta = 3$, synthesis fuel contains excess steam, the production of which requires heat. Therefore, with temperature increasing after 700°C, a decrease in the recovery rate for $\beta > 1$ is more significant than for $\beta = 1$.

Fig. 7 makes it clear that, at a temperature of 700°C, the TCR system is capable of recovering more than 85% heat of exhaust gases after a gas furnace.

Figs. 6 and 7 shows the dashed line for $\beta = 3$. The dashed line shows the total recovered heat and the heat recovery rate by taking into account the heat deficit. Fig. 8 shows the enthalpy of cold exhaust gases after the steam generator (diagram Fig. 1) versus

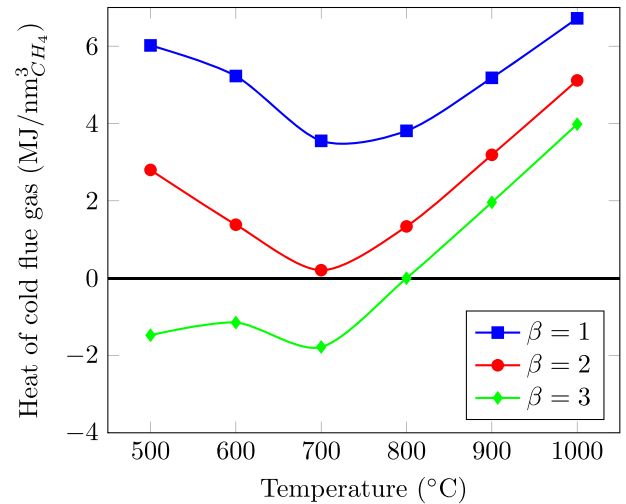


Fig. 8. The enthalpy of cold exhaust gases after the steam generator (diagram Fig.1) versus temperature of exhaust gases after a gas furnace.

temperature of exhaust gases after a gas furnace.

Fig. 8 that with an increase in the steam-to-methane ratio, the heat input for the steam production increases. When $\beta = 3$, the enthalpy of the exhaust gases after the gas furnace is not enough to perform thermochemical recovery with the initial data shown in Table 2. Therefore, there is a heat deficit in the TCR system. Such the heat deficit can be filled by the supply of heat from the furnace. Fig. 8 shows that if the enthalpy of cold exhaust gases is negative, then this means that there is a heat deficit in the TCR system. Therefore, the recovery rate with taking into account the heat deficit for $\beta = 3$ is lower than the recovery rate for $\beta = 2$.

In addition, Fig. 8 shows that in the temperature range of exhaust gases after the furnace above 700°C the cold exhaust gases after the steam generator has residual heat. This residual heat can be used to heat the air for synfuel combustion. Air heating simultaneously with thermochemical recovery is a promising way of an efficiency increase of the gas furnaces.

5. Conclusion

This paper considers an industrial gas furnace with a thermochemical waste-heat recuperation system based on the steam methane reforming process. The main criterion of efficiency of the recuperation systems is the heat recovery rate – the ratio between a recovered heat to an exhaust heat after a gas furnace.

The methodology for determining the recovery rate is developed in this paper. This methodology is based on determining the recovered heat in a reformer and a steam generator using experimental data on synthesis gas composition. Based on the experimental results and the developed methodology, the following results were obtained:

- a recovered heat increases with increasing exhaust gas temperature; in the temperature range up to 700°C, an increase in a recovered heat takes place mainly due to an increase in the enthalpy of the steam methane reforming process;
- the heat recovery rate for the TCR systems with a steam-to-methane ratio from 1 to 3 has a maximum value in the temperature range of 700–800°C;
- the maximum heat recovery rate 0.852 is achieved at a temperature of about 700°C and a H₂O:CH₄ ratio of 2;

- if the exhaust gas temperature after a furnace is less than 800°C, then there is a heat deficit in the TCR system with a steam-to-methane ratio of 3; therefore, external heat supply is necessary;
- the heat deficit is about $2 \text{ MJ/m}^3_{\text{CH}_4, \text{in}}$ for the TCR system with steam-to-methane ratio of 3 and the exhaust gas temperature of 700°C.

In the temperature range of exhaust gases after the furnace above 700°C the cold exhaust gases after the steam generator have residual heat. This residual heat can be used to heat the air for synfuel combustion. Air heating simultaneously with thermochemical recovery is a promising way of an efficiency increase of the gas furnaces which have to be investigated by the engineers and scientists.

Credit author statement

The article was written by one author.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2020.117395>.

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